

Gas-vapor mixtures and Air-Conditioning

Mixture of gases (mass fractions, mole fractions,
P-V-T, properties)

↳ Human comfort
↳ basic applications
for Eng. Mech. & Act.

↳ Components ^(remained) were in gas phase



Now one of the component is to change phase

Gas-vapor mixture → Dry air & water vapor

• Air-conditioning

• Atmospheric sciences

- cloud formation & convection

- Short & long-wave radiative flux

- Climate change

Dry Air → Mixture of
 N_2, O_2 , small amount of other gases)

↳ No water

Atmospheric air - mixture of dry air & water vapor

How to analyze

Recall Dalton's Law: $P_m = \sum_{i=1}^k P_i (@ V_m, T_m)$

↳ led to partial pressures

$$P_i = P_m Y_i$$

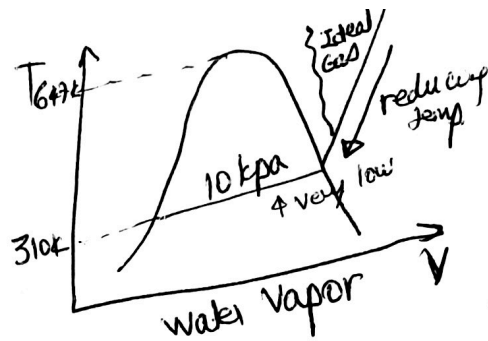
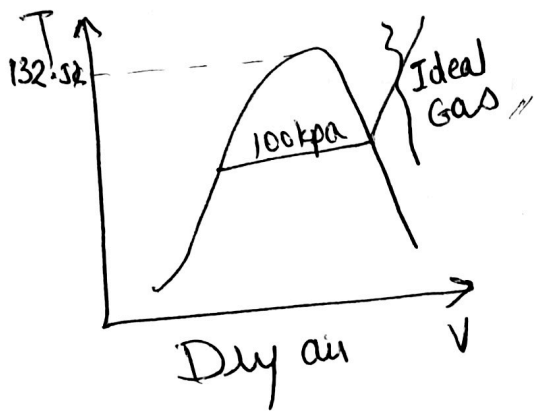
⇒ Atmospheric air:

$$P = P_a + P_v$$

↑
Partial pressure
for dry air

Partial pressure water vapor
(vapor pressure)

Both treat as ideal gas &
we are most interested in temperature range
b/w -10°C to 50°C,



For atmospheric air: lower temp. at fixed pressure
 ↳ you hit the vapor saturation line
 ↳ dew
 fog
 rain
 snow

Change pressure then
 ↳ Saturation temp. (dewpoint)
 goes down.

④ Airplane
 Different regions

$$T_{dp} = T_{sat}@P_v$$

Develop E_{gns} to solve
Enthalpy Approximation (-10°C → 50°C)

Dry Air: $dh = C_p dT$
 $\int_{0^\circ C}^T dh = C_p \int_{0^\circ C}^T dT$

$$h(T) - h(0^\circ C) = C_p(T - 0)$$

let 0°C be the reference state, $h(0^\circ C) = 0$

$$h(T)_{dry air} = 1.005 \frac{kJ}{kg \cdot ^\circ C} (T) \quad \text{Temp is in } ^\circ C$$

$$\Delta h_{dry air} = 1.005 \Delta T \left(\frac{kJ}{kg} \right)$$

Water Vapor:

$$h_v(T, \text{low } p) \approx h_g(T) \quad \text{Saturated vapor enthalpy at the given temp.}$$

h_g can be approximated as:

$$h_g(T) \approx 2501.3 + 1.82 T \quad \left(\frac{\text{kJ}}{\text{kg}} \right)$$

$\uparrow$ in $^{\circ}\text{C}$

Amount of ~~humidity~~ (water vapor) the air actually can hold:

↳ Specific Humidity (w)

$$w = \frac{m_v}{m_a} = \frac{\text{water vapor mass}}{\text{Dry air mass}}$$

Using the Ideal Gas Law/Eqn ($pV = mRT$):

$$w = \frac{p_v V_v / R_v T_v}{p_a V_a / R_a T_a} = \frac{R_a}{R_v} \frac{p_v}{p_a} \quad (@T, V)$$

$\frac{0.287}{0.4615}$

$$w = 0.622 \frac{p_v}{p_a}$$

And $p = p_a + p_v$

$\Rightarrow$

$$w = 0.622 \frac{p_v}{p - p_v}$$

$\therefore$ We can keep adding water vapor up until we reach the saturated air point.

↳ air will not take any more moisture.
↳ Condensation starts.

→ Amount of humidity b/n dry air & saturated air:
 $P_v |_{\text{sat air}} = P_g = P_{\text{sat pressure}}$ at temperature T

$$\omega |_{\text{sat air}} = 0.622 \frac{P_g}{P - P_g}$$

* How much moisture we can add? (limit)
 $P_v = P_g$

↳ Relative Humidity (ϕ)

$$\phi = \frac{m_v}{m_g} = \frac{\text{Moisture in air}}{\text{Maximum moisture air can hold}}$$

Using Ideal Gas Law/Eqn ($PV = mRT$)

$$\phi = \frac{P_v V / R_v T}{P_g V / R_v T} = \frac{P_v}{P_g} \quad (@ V \& T)$$

* $P_g = P_{\text{sat}} @ T$ from steam tables.

* Combine ϕ and ω

$$\phi = \frac{\omega P}{(0.622 + \omega) P_g} ; \quad \omega = \frac{0.622 \phi P_g}{P - \phi P_g}$$

→ Enthalpy (per unit mass dry air):
 $H = H_a + H_v = m_a h_a + m_v h_v$
 divide by m_a : $h = h_a + \frac{m_v}{m_a} h_v$
 mass of dry air is not changing
 ω

$$h = h_a + \omega h_v$$

Enthalpy per kg of dry air.

→ Change in Enthalpy $h_2 - h_1$:

$$h_2 = h_{a2} + \omega_2 h_{v2}$$

$$h_1 = h_{a1} + \omega_1 h_{v1}$$

$$\Rightarrow h_2 - h_1 = \underbrace{h_{a2} - h_{a1}}_{\text{dry air}} + \underbrace{\omega_2 h_{v2} - \omega_1 h_{v1}}_{\text{water vapor}}$$

→ Dry air

$$\rightarrow h_{a2} - h_{a1} = C_p T_2 + h(0^\circ\text{C}) - C_p T_1 - h(0^\circ\text{C})$$

$$C_p(T_2 - T_1)$$

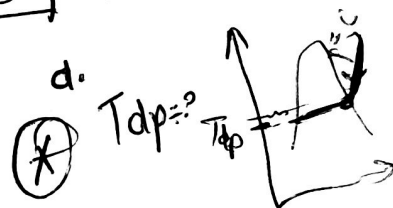
→ For water vapor: ^{not equal}
as $\omega_2 \neq \omega_1$

$$\Rightarrow h(0^\circ\text{C}) = 2501.3$$

Example

A tank contains 21 kg of dry air and 0.3 kg of water vapor at 30°C & 100 kPa total pressure. Determine:

- the specific humidity
- the relative humidity
- the volume of the tank.



Given

21 kg - dry air
0.3 kg - water vapor
 $T = 30^\circ\text{C}$
 $P = 100\text{ kPa}$

Assumption

Both dry air & water vapor
treated as ideal gas.

$$P_v = 0.5288 \cdot P_{\text{sat}} T$$

$$P_a = P - P_v = 99.755\text{ kPa}$$

$$v = \frac{m_a R_a T}{P_a}$$

$$v = \frac{(21)(0.287)(303)}{99.755}$$

$$V = 18.68\text{ m}^3$$

Analysis

$$a. \omega = \frac{m_v}{m_a} = \frac{0.3}{21} = 0.0143$$

$$b. \phi = \frac{\omega P}{(0.622 + \omega) P_g}$$

$$\text{And } P_g = P_{\text{sat}} @ 30^\circ\text{C} = 4.246\text{ kPa (steam tables)}$$

$$\phi = \frac{(0.0143)(100)}{(0.622 + 0.0143)(4.246)}$$

$$= 0.5288$$

$$c. P_a V = m_a R_a T \quad \text{and} \quad P_a = P - P_v$$

$$P - P_{\text{sat}} T = \phi$$

$$\Rightarrow T_{dp} = T_{\text{sat}} @ P_v$$

Dry-Bulb / Wet-Bulb Temperature humidity using 2 thermometers.

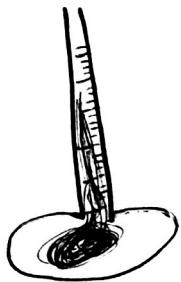
→ Measure relative

↳ Dry bulb thermometer

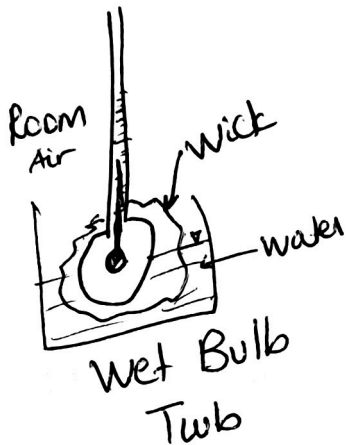
↳ measures standard atmospheric condition.

↳ Wet bulb thermometer

↳ measures 100% relative humidity (saturated air.)



Room Air
Dry Bulb
 T_{db}



⇒ Enable us to determine ϕ

State ① - Dry Bulb

$$\phi_1 = ?$$

$$T_1 = T_{db}$$

State ② - Wet Bulb

$$\phi_2 = 100\%$$

$$T_2 = T_{wb}$$

$$\Rightarrow \omega_1 = \frac{c_p(T_2 - T_1) + \omega_2 h_{f2}}{h_{g1} - h_{f2}}; \quad \omega_2 = \frac{0.622 P_{g2}}{P - P_{g2}}$$

$$\phi_1 = \frac{\omega_1 P}{(0.622 + \omega_1) P_{g1}}$$

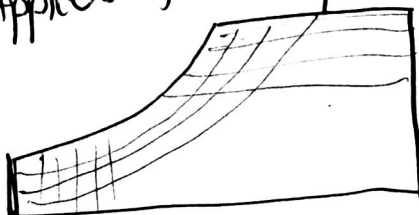
To Ease calculations:

Shortcut:

Psychrometric chart:

• Determine many properties:

↳ Applies for a specific atmospheric pressure



1 atm
2 atm

Example

The air in a room is at 1 atm , 32°C , and 60% relative humidity. Using the psychrometric chart, determine:

- Specific humidity
- Enthalpy (kJ/kg dry air)
- Wet bulb temperature
- Dew point temperature
- Specific volume ($\text{m}^3/\text{kg of dry air}$)

Given

$\rightarrow 1 \text{ atm} \rightarrow$ we can use psychrometric chart:

Required

- ω
- h
- T_{wb}
- T_{dp}
- v

$$\Rightarrow \omega = 18.9 / \text{kg d.a.} = 0.018$$

$$h = 79 \text{ kJ} / \text{kg d.a.}$$

$$T_{wb} = 25.5^\circ\text{C}$$

$$T_{dp} = 23^\circ\text{C}$$

$$v = 0.89 \text{ m}^3 / \text{kg d.a.}$$

Air-Conditioning Process

$\hookrightarrow$ HVAC $\Rightarrow$ Heating Ventilation, and Air-Conditioning

$\hookrightarrow$ Simple Heating

$\hookrightarrow$ Simple Cooling

$\hookrightarrow$ Humidifying

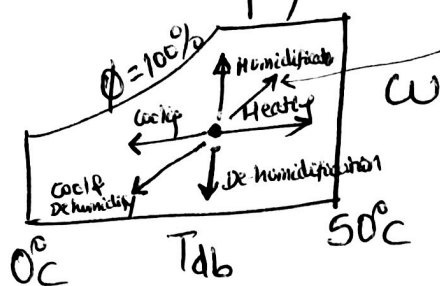
$\Rightarrow$ Adding moisture; reduce dryness

$\hookrightarrow$ Dehumidifying

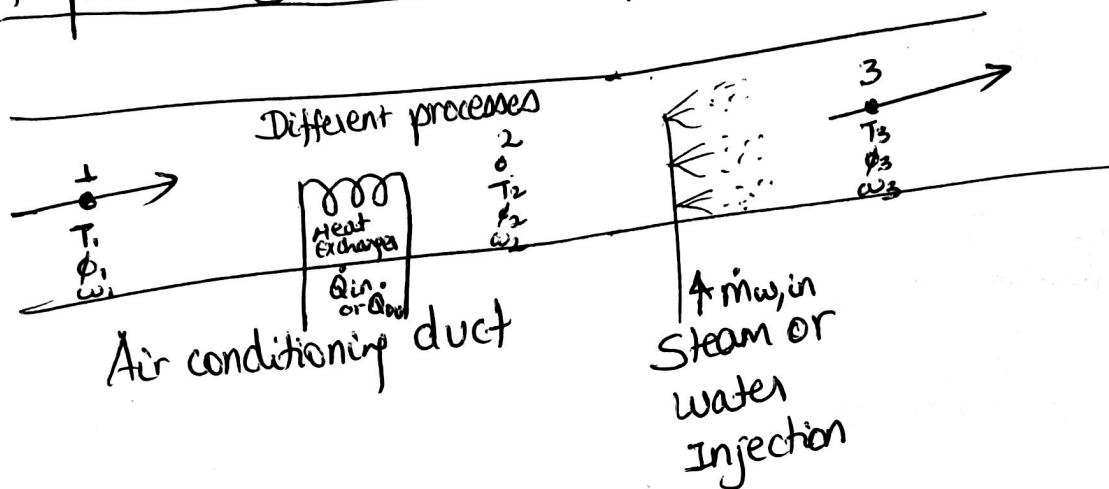
And combinations of these

(*) If the HVAC process takes place at 1 atm , then we can use the psychrometric chart and the analysis is rather quick. If the process is not at 1 atm & we do not have an altitude-correlated chart, we have to use the equations.

Recall the psychrometric Chart:



A process schematic usually looks like:



Conservation Equations:

→ Mass
→ Energy

Mass Balance Air:

$$\sum \dot{m}_{air(i)} = \sum \dot{m}_{air(e)}$$

$$\dot{m}_{a,1} = \dot{m}_{a,2} = \dot{m}_{a,3}$$

Mass Balance Water:

$$\sum \dot{m}_{w,i} = \sum \dot{m}_{w,e}$$

$$\dot{m}_{w,1} = \dot{m}_{w,2} \quad \text{if Heating or cooling}$$

$$\dot{m}_{w,1} + \dot{m}_{w,in} = \dot{m}_{w,3}$$

and:

$$\dot{m}_{w,i} = \omega_i \dot{m}_{a,i}$$

$$\frac{\text{kg H}_2\text{O}}{\text{kg of dry air}}$$

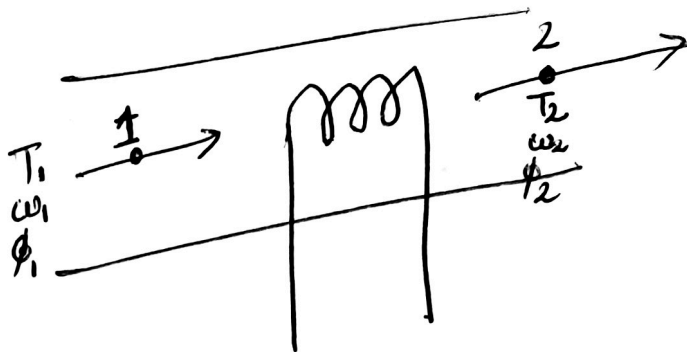
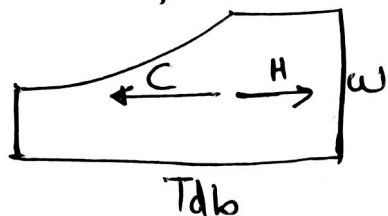
Energy conservation

First Law:

$\dot{Q}_{in} + \dot{W}_{in} + \sum \dot{m}_i h_i = \dot{Q}_{out} + \dot{W}_{out} + \sum \dot{m}_e h_e$
Enthalpy from psychrometric chart or use approximation Eq: $h_i = h_{a,i} + \omega_i h_{o,i}$

$$h_i = 1.005 T_i + \omega_i (2501.3 + 1.82 T_i)$$

1. Heating and Cooling



Air: $\dot{m}_{a,1} = \dot{m}_{a,2}$

Water: $\dot{m}_{w,1} = \dot{m}_{w,2} \rightarrow$ simple H or C
 $w_1 \dot{m}_{a,1} = w_2 \dot{m}_{a,2}$ And $w = \frac{kg H_2O}{kg \text{ of dry air}}$

$\Rightarrow w_1 = w_2$

Energy: 1st Law: $\dot{Q}_{in} + \dot{W}_{in} + \sum \dot{m}_i h_i = \dot{Q}_{out} + \dot{W}_{out} + \sum \dot{m}_e h_e$
 Let we assume there is heat addition, with single inlet & exit.

$\Rightarrow \dot{Q}_{in} = \dot{m}_e h_e - \dot{m}_i h_i$

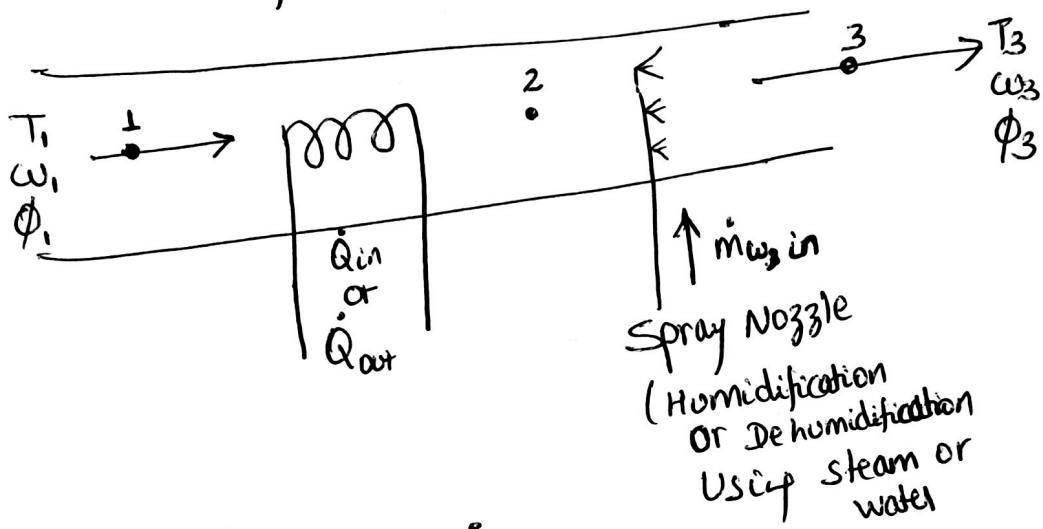
$= \dot{m}_a (h_e - h_i)$

$h_2 = h_{a2} + w_2 h_{v2} \left[\frac{kJ}{kg \text{ of dry air}} \right]$

or

Psychrometric chart
 (if $p = 1 \text{ atm}$ or other given)

2. Heating and Cooling with Humidification / Dehumidification.



Mass

$$\begin{cases} \text{Air: } \dot{m}_{a,1} = \dot{m}_{a,2} = \dot{m}_{a,3} \\ \text{Water: } \dot{m}_{w,1} + \dot{m}_{w,in} = \dot{m}_{w,3} \end{cases}$$

$$\dot{m}_a \omega_1 + \dot{m}_{w,in} = \dot{m}_a \omega_3$$

Energy: 1st Law

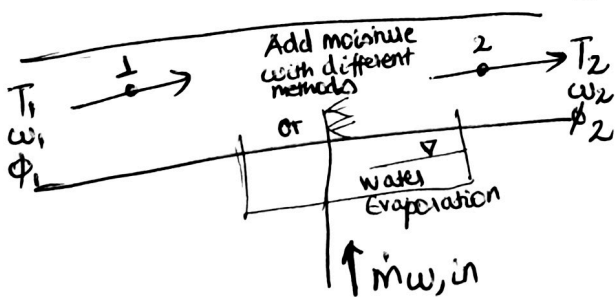
$$\dot{Q}_{in} + \underbrace{\sum \dot{m}_i h_i}_{\text{air \& steam}} = \underbrace{\sum \dot{m}_e h_e}_{\text{Air}}$$

$h_{\text{water,in}}$ from steam tables = $h_f @ T$ ↳ saturated liquid

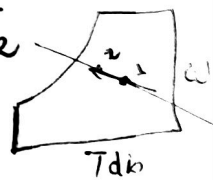
3. Evaporative cooler (Swamp cooler)

↳ for dry air (Desert)

↳ Since it humidifies will not work for humid air



Usually assume $T_{w,in} = T_2$
In this process, $h_1 \approx h_2$

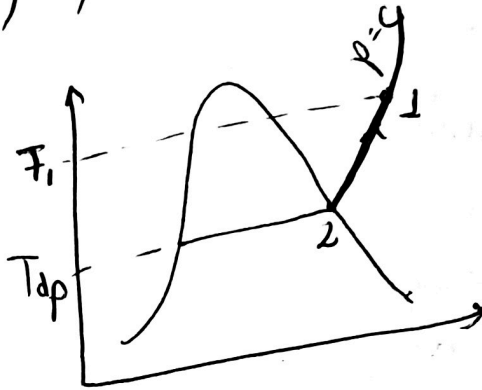


Example

$$T = 25^\circ\text{C}, p = 100 \text{ kPa}$$

$$V = 150 \text{ m}^3, \phi = 60\%$$

T_{dp} :



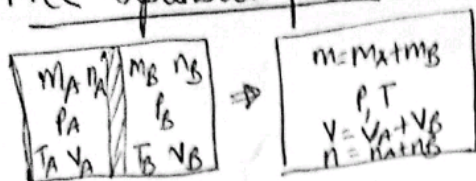
⊗ Temp @ which condensation start following constant pressure.

$$P_v = ?$$

$$T_{dp} = T_{sat} @ P_v$$

Adiabatic mixing of perfect gases

Free expansion process



Internal energy initially = Internal energy finally

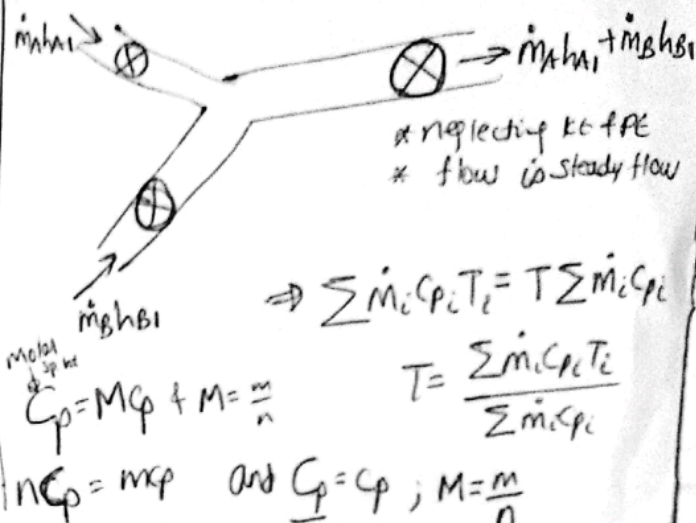
$$U_1 = n_A C_{vA} T_A + n_B C_{vB} T_B = U_2 = (n_A C_{vA} + n_B C_{vB}) T$$

$$U_1 = \sum n_i C_{vi} T_i = U_2 = T \sum n_i C_{vi}$$

$$\sum n_i C_{vi} T_i = T \sum n_i C_{vi}$$

$$T = \frac{\sum n_i C_{vi} T_i}{\sum n_i C_{vi}}$$

Two streams of fluid meet to form a common stream.



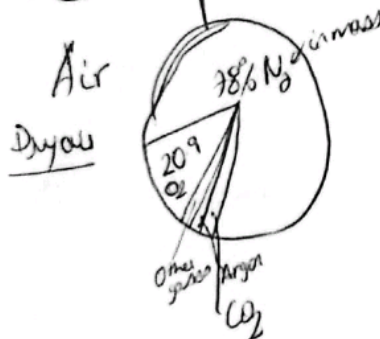
$$\sum \dot{m}_i C_{pi} T_i = T \sum \dot{m}_i C_{pi}$$

$$T = \frac{\sum \dot{m}_i C_{pi} T_i}{\sum \dot{m}_i C_{pi}}$$

$$C_p = M C_p + M = \frac{m}{n}$$

$$n C_p = m C_p \text{ and } C_p = C_p; M = \frac{m}{n}$$

2. Gas vapor mixture



Air + water vapor

Mix of ideal gases

What to pick: Amagat or Dalton

Dalton's model: Air + water vapor is everywhere accompanying the whole volume

Partial pressure of water vapor: P_v

Partial pressure of dry air: $P_{dry air}$

Enthalpy of saturated water at same temp: $h_v = h_g(T)$

$$T = \frac{\sum n_i C_{pi} T_i}{\sum n_i C_{pi}}$$

Restraining ourselves to Atmospheric pressure

Feels like $-5^\circ C$ due to the wind

Humidity 93%

Dew pt temp $-1^\circ C$

The air is almost full @ 93% of its capacity

Humidity 8%

Dew $14^\circ C$

The air is very far from its max. capacity

Winds chose to dry air then with saturated air.

Relative humidity

$$\phi = \frac{m_v}{m_g} = \frac{\text{mass of water vapor currently included in our air}}{\text{maximum water vapor the air can hold}}$$

$$= \frac{P_v \sqrt{1/\epsilon} T}{P_g \sqrt{1/\epsilon} T}$$

$$\phi = \frac{P_v}{P_g^{(sat)} \text{ at the temp } T}$$

$$0 < \phi < 1$$

0% < ϕ < 100% from table

to be compared to a certain value

Humidity ratio (w); Specific humidity

$$w = \frac{m_v}{m_a} = \frac{\text{mass of water vapor}}{\text{mass of dry air}}$$

The water vapor is dictated by its partial pressure

In this room mass of water vapor what we hv compared to the rest of dry air without water vapor.

$$= \frac{P_v \sqrt{1/\epsilon} T}{P_a \sqrt{1/\epsilon} T} = \frac{P_v}{P_a} \cdot \frac{P_v = 0.018 P_a}{0.4615 P_a} \cdot \frac{P_v}{0.622 P_a}$$

$$W = \frac{m_v}{m_a} = \frac{\frac{P_v V_v}{R_v T_v}}{\frac{P_a V_a}{R_a T_a}}$$

Since $V_v = V_a$ & $T_v = T_a$

$$W = \frac{P_v}{P_a} \cdot \frac{R_a}{R_v}$$

$$= \frac{287.1 \text{ J/kg} \cdot \text{K}}{461.4 \text{ J/kg} \cdot \text{K}} \cdot \frac{P_v}{P_a}$$

$$= 0.622 \frac{P_v}{P - P_a} = 0.622 \frac{P_v}{P - P_a}$$

$PV = nRT$
 $PV = mRT$
 $m = \frac{PV}{RT}$
 Ideal gas eqn:
 Dalton's Model:
 @ (T, V)

$$R_a = 287.1 \text{ J/kg} \cdot \text{K}$$

$$R_v = 461.4 \text{ J/kg} \cdot \text{K}$$

$$\phi = \frac{m_v}{m_g} = \frac{P_v V / RT}{P_g V / RT} = \frac{P_v}{P_g}$$

$$P_g = P_{\text{sat}} @ T$$

P_g is a function of T
 $P_g = f(T)$
 The saturated pressure increases under the temperature

Combining above eqn with air.

$$\phi W = 0.622 \frac{P_v}{P - P_v} = 0.622 \frac{P_g}{P - P_g}$$

① Enthalpy of air:

$$H = h_a + H_v = m_a h_a + m_v h_v$$

Specific enthalpy:

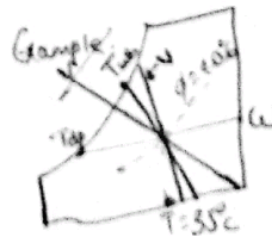
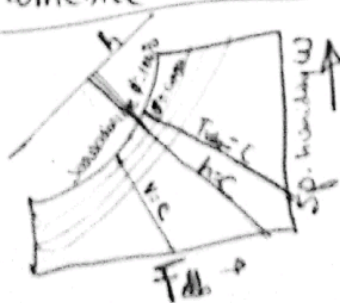
$$h = h_a + \frac{m_v}{m_a} h_v = h_a + W h_v$$

$$T = 0^\circ \text{C reference}$$

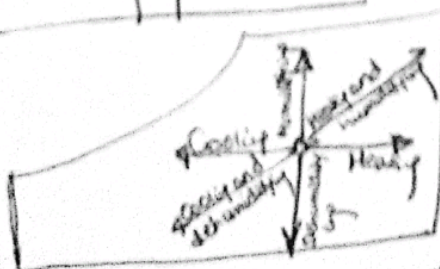
$$h_a =$$

cooling tower can be by distillation

→ The psychrometric chart

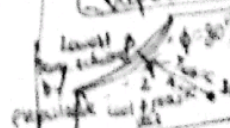


→ Air conditioning process



Evaporative cooling

- Essentially identical to adiabatic saturation process.



Adiabatic Mixing

- State of mixture lies on straight line connecting states 1 & 2

Wet cooling tower

- Water sprays through pressurized nozzles near the top of the tower & then flows downward through the fill, opposite to the air flow.

$$\Rightarrow W = 0.622 \frac{P_v}{P_a}$$

$$= 0.622 \frac{P_v}{P - P_v}$$

$$\text{And } \phi = \frac{P_v}{P_g}$$

$$\Rightarrow P_v = \phi P_g$$

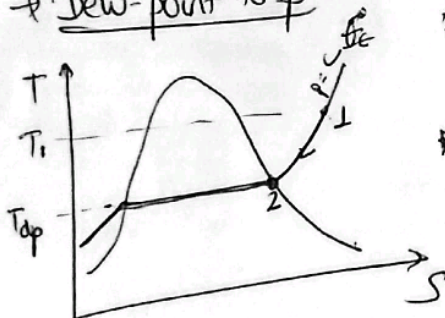
$$\Rightarrow W = 0.622 \frac{\phi P_g(T)}{P_a}$$

→ Dry bulb temp

Temp read by thermometer

Temp doesn't consider wind effect or presence of humidity.

→ Dew-point temp



$$\Rightarrow 1^\circ C \leftarrow T_{dp} \leftarrow 0^\circ C$$

$$\Rightarrow 14^\circ C \leftarrow 31^\circ C$$

From 31 to 14

we will see water droplets

⊛ Simple fluid, pure substance

⊛ 2 independent intensive properties → give state

But for mixture we need 1 extra property

Mixture of air & water vapor

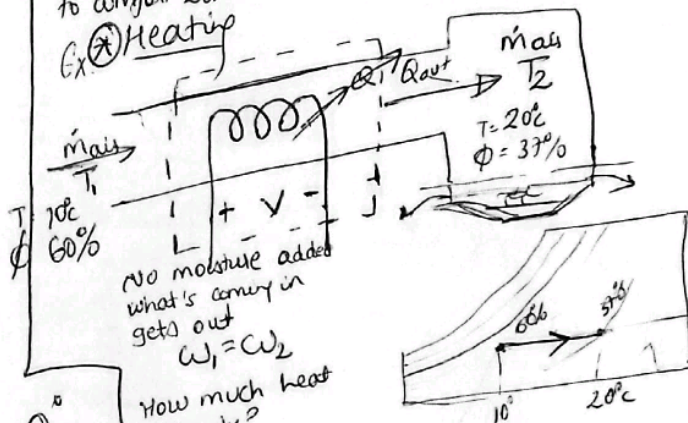
T, P & w

Assume P = 1 atm always

T, w, φ, w

How to bring any given condition to comfort zone

Ex: Heating



How much heat to supply?

Energy balance in the system

$$\frac{dE}{dt} = \dot{Q}_{in} - \dot{W}_{in} + \dot{m}_{air}(h_1 - h_2)$$

$$0 = \dot{Q}_{out} + \dot{m}_{air}(h_1 - h_2)$$

$$\frac{\dot{Q}_{out}}{\dot{m}_{air}} = \frac{\dot{m}_{air}(h_2 - h_1)}{\dot{m}_{air}} = h_2 - h_1$$

$$h_2 \text{ \& } h_1 \text{ from data}$$

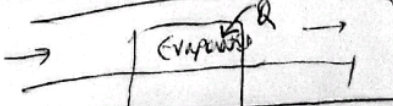
$$h_2 = 32.5$$

$$h_1 = 22$$

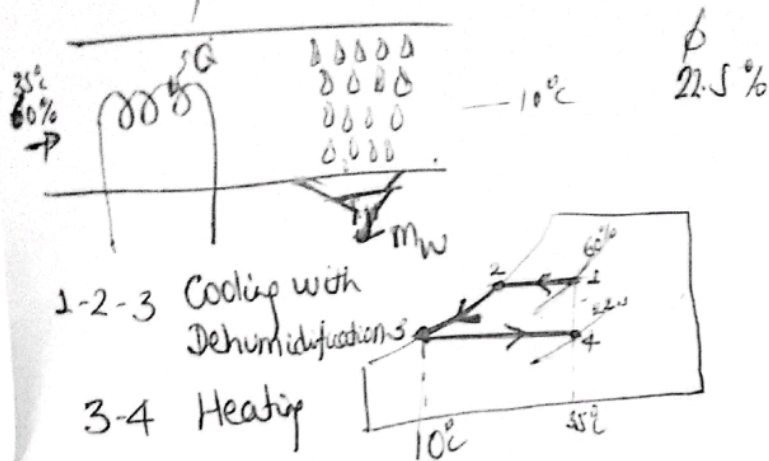
possible to size heaters, how many, where ---

⊛ Cooling

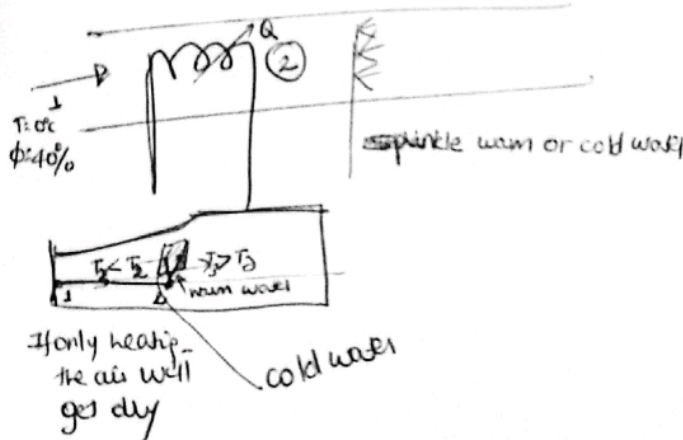
The same thing except



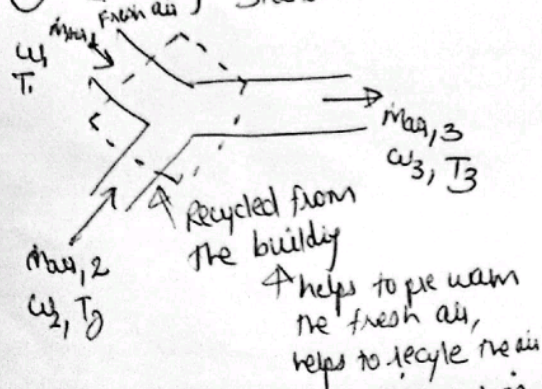
⊗ Cooling and De-humidification



⊗ Heating with Humidification



⊗ Mixing standard construction in buildings



- Assumptions
- steady flow
- insulated
- no work
- $\Delta KE + \Delta PE \approx 0$

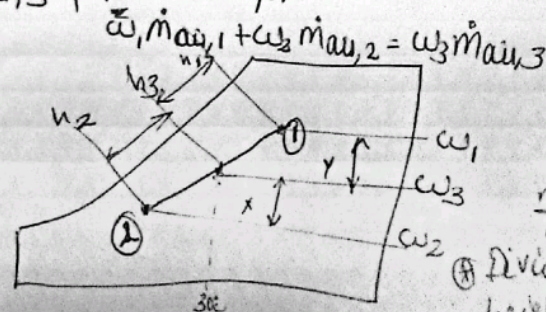
Conservation of mass: $\dot{m}_{a1} + \dot{m}_{a2} = \dot{m}_{a3}$ for water vapor

Conservation of energy: $\dot{m}_{a1}h_1 + \dot{m}_{a2}h_2 = \dot{m}_{a3}h_3$

$$\dot{m}_{a1}h_1 + \dot{m}_{a2}h_2 = \dot{m}_{a3}h_3$$

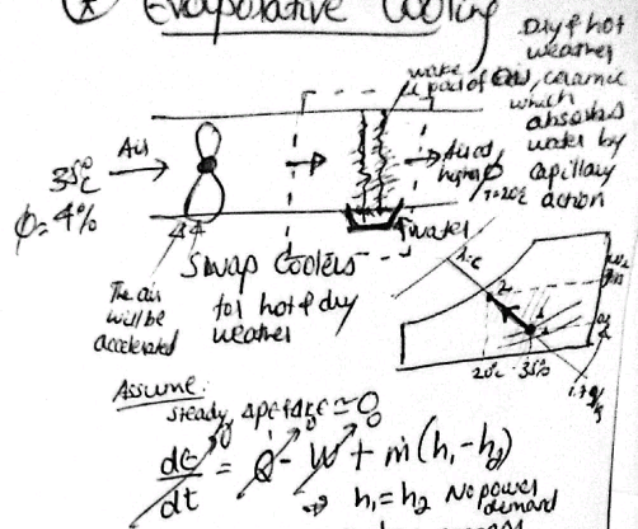
Eliminate $\dot{m}_{a3}$

$$\frac{\dot{m}_{a1}}{\dot{m}_{a2}} = \frac{h_3 - h_2}{h_1 - h_2} = \frac{w_3 - w_2}{w_1 - w_2}$$



Divide the heights: x measured by ruler

⊗ Evaporative Cooling



Assume: steady state $\dot{Q} = 0$

$\frac{d\dot{Q}}{dt} = \dot{Q} - \dot{W} + \dot{m}(h_1 - h_2)$

$\dot{h}_1 = \dot{h}_2$ No power demand

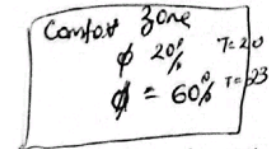
⊗ Constant enthalpy process

$\dot{m} = \text{constant}$ that is why its good for dry & hot weather

$h_1 = h_2$

$\dot{m}_{\text{liquid water}} = (w_2 - w_1) \dot{m}_{\text{air}}$

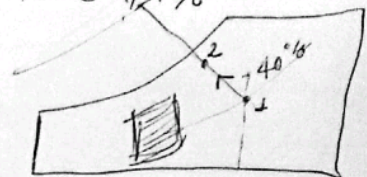
To size Speed of water consuming water to cool this place



possible to attain comfort zone using evaporative cooling

Hot & humid

$T = 35^\circ\text{C}$ $\phi = 70\%$



Impossible to get to comfort zone using ev. cooling for hot & humid